

Dean Foulds

Data Scientist & ML Engineer

deanfoulds@gmail.com · deanfoulds.xyz · linkedin.com/in/dean-foulds · github.com/Dean-Foulds

PERSONAL STATEMENT

I approach data science as a problem solver first — applying mathematical rigour and scientific methodology to untangle complex, high-stakes systems. My career spans over a decade across aerospace, motorsport, and subsea infrastructure: environments where disciplined thinking and verifiable results are not optional, and where the gap between a good model and a correct one genuinely matters.

I hold a BSc (Hons) in Mathematics and Physics from the Open University and an Airline Transport Pilot's Licence — qualifications that reflect a consistent drive toward precision in high-stakes environments. I specialise in the rapid deployment of agentic AI systems, JAX-based ML pipelines, and formal proof verification in Lean 4 and Mathlib, with over 30 machine-verified proofs currently applied to live production aerospace risk models alongside GARCH and MCMC systems. I have taped out two chips in silicon via Tiny Tapeout, spanning binary neural network inference and systolic accelerator design. I consistently deliver functional autonomous systems within 8–12 weeks, typically achieving ROI improvements of 15–30% above projections — not through shortcuts, but through the kind of mathematical clarity that makes complex systems tractable.

TECHNICAL SKILLS

CATEGORY	SKILLS
Languages & Frameworks	Python · Lean 4 / Mathlib · JAX / NumPyro · TensorFlow · PyTorch · Polars · Verilog / HDL · LaTeX · Ruby
Cloud & Infrastructure	AWS SageMaker · Lambda · EMR · Google Vertex AI · BigQuery · Cloud Functions · Docker · Terraform
ML & Mathematics	MCMC Modelling · GARCH · Monte Carlo · Statistical Learning · Agentic AI · LangChain / LangGraph · SPC
Engineering	Six Sigma DMAIC · FMEA / PPAP · Lean Manufacturing · A/B Testing · Agile / Scrum · CAD (Fusion 360 / Creo)

NOTABLE PROJECTS

Risk Algorithm - Aircraft Parts · AJW Aerospace / Personal Portfolio · 2025–Present

- Production ML pipeline combining GARCH volatility modelling and MCMC inference (Python, JAX) for aircraft parts pricing, with core mathematical proofs verified in Lean 4 and Mathlib — covering volatility theory, pricing guardrails, and market propagation via the Banach fixed-point theorem.
- Demonstrates probabilistic inference paired with machine-checked correctness in a live commercial system. Proofs at deanfoulds.xyz/projects/lean_projects.

Lean 4 Theorem Proving · Personal Portfolio · 2024–Present

- Over 30 machine-verified proofs in Lean 4 and Mathlib spanning number theory, real analysis, algebra, and combinatorics — co-located with production engineering projects as correctness guarantees rather than academic exercises.

Systolic BNN Accelerator (v2) & 16-Neuron BNN · Tiny Tapeout · 2024–2025

- Two chips taped out in silicon: a 16-neuron BNN achieving single-cycle 8-bit inference with no CPU or software, and a redesigned v2 systolic accelerator with XNOR dot product engine, signed bias, and balanced popcount tree.

MCMC Model Training Pipeline · Personal Portfolio · 2023–Present

- Full MCMC model training pipeline with continuous retraining, visualisations, and predictions-vs-actuals analysis — built to production standard with reproducible results and systematic validation.

AMP Statistical Process Control · Personal Portfolio · 2023

- ML-driven SPC system for predictive analytics and process monitoring, applying statistical learning theory to detect drift and anomalies before they propagate to defects.

DSF Talk Explorer · Personal Portfolio · 2024–Present

- Agentic LLM application powered by a Neo4j graph database for natural language querying of Data Science Festival London talks — speaker, topic, and event data structured as a knowledge graph with a mobile-first interface.

Bioinformatics - Squamous Cell Carcinoma · Personal Portfolio · 2023

- Computational analysis of squamous cell carcinoma using Python and Jupyter, applying data science methodology to genomic and clinical data to surface statistically significant patterns.

WORK EXPERIENCE

Data Scientist · AJW Aerospace · Apr 2025 – Present · Sussex

- Designed and deployed autonomous agentic AI systems within 8–12 week cycles, consistently achieving ROI improvements of 15–30% above projections.
- Architected end-to-end ML pipelines on AWS and Google Cloud; developed Lean 4 / Mathlib proofs co-located with production risk models, providing machine-verified correctness on live pricing algorithms.
- Applied Monte Carlo simulation, DCF analysis, MCMC modelling, and decision theory frameworks for risk assessment and strategic decision-making.

Data Scientist & ML Engineer · ASN Submarine Cables · Dec 2023 – Apr 2025 · London

- Converted corrective to predictive maintenance using ML failure prediction and machine vision anomaly detection, measurably reducing unplanned downtime across fibre optic and laser manufacturing cells.
- Applied SPC and MCMC-based predictive models to reduce defect escape rates; led Kaizen events improving cell KPIs measured by DPMO via DMAIC and 8D problem-solving.

Complex Data Analysis · McLaren Racing · Mar 2020 – Dec 2021 · Woking

- Implemented SPC and predictive ML models to reduce manufacturing defects; led FMEA, PPAP, and A/B testing protocols across vehicle production lines using Python, MATLAB, and Octave.

Aeronautical Engineering Lecturer (Contract) · East Surrey College · Mar 2022 – Dec 2023 · Redhill

- Delivered mathematics, physics, Python, CAD, and Six Sigma DMAIC; designed hands-on VSM and SPC exercises for aerospace engineering students.

Quality Engineer · PGT Ceewrite Precision · May 2015 – Dec 2019 · Mitcham, Surrey

- Document and tooling design approvals; customer representative during audits maintaining TL9000 compliance. DMAIC and 8D root cause analysis driving sustained process improvements and reduced NPI lead times.

Quality Engineer & Mechanical Inspector · Ametek Aerospace · Oct 2013 – Mar 2015 · London

- EASA Part 145/21 and FAR Part 145 compliance; CMM inspections of aerospace and medical instruments, zero non-conformance escapes to customer.

Contract Project Manager · Eaton Aerospace · Feb 2007 – Apr 2010 · Fareham

- Managed £5M+ programmes (Eurofighter Typhoon, Tornado); used earned value techniques to convert a two-month delay into a two-month early delivery. Led teams of up to 30.

First Officer / Flight Test Engineer · Skybus / BASCO · 2006 – 2013 · Cornwall & Bournemouth

- DHC6 island discipline First Officer; NPI lead on military sensor manufacturing project overseeing a team of 10 and engineering group of 20+.

EDUCATION & CERTIFICATIONS

QUALIFICATION	INSTITUTION	YEAR
BSc (Hons) Mathematics and Physics	Open University	2018
Professional Machine Learning Engineer	Google	2024
Quantum Information	IBM	2024
Machine Learning	Stanford University (Coursera)	2020
ATPL & Flight Instructor Rating	BCFT	2006
Prince2 Agile · PMI CAPM	Learning People	2013

Languages: English (native) · French · German · **Portfolio:** deanfoulds.xyz · **GitHub:** github.com/Dean-Foulds